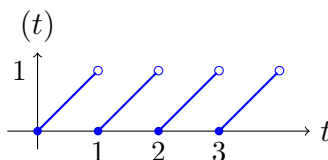


Explainer: Problem 4 (Rudin 7.10) — Fractional Part Series

What's going on?

The function is $f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2}$, where $(t) = t - [t]$ is the **fractional part** of t .



The fractional part (t) is a sawtooth wave: it climbs linearly from 0 to 1, then jumps back to 0 at every integer. It's continuous except at integers, where it has a jump discontinuity.

Where is (nx) discontinuous?

The function $t \mapsto (t)$ is discontinuous at integers. So $x \mapsto (nx)$ is discontinuous when nx is an integer, i.e. when x is a rational of the form $\frac{k}{n}$.

The n th term $\frac{(nx)}{n^2}$ is discontinuous at $x = \frac{k}{n}$ for $k \in \mathbb{Z}$.

Where is f discontinuous?

At every rational $x = p/q$: The term with $n = q$ (and multiples of q) is discontinuous there.

At irrationals: Every single term $(nx)/n^2$ is continuous (since nx is never an integer when x is irrational). The series converges uniformly (by the M -test, since $|(nx)/n^2| \leq 1/n^2$ and $\sum 1/n^2 < \infty$), so the sum of continuous functions is continuous.

Wait — but at rationals, the series also converges uniformly! So why isn't f continuous at rationals?

The subtlety: At $x = p/q$, some of the terms themselves are discontinuous. Theorem 7.12 says “uniform limit of continuous functions is continuous.” If the individual terms aren't all continuous at x_0 , the theorem doesn't apply.

Proving f is actually discontinuous at rationals

At $x = p/q$ (in lowest terms), the term $(qx)/q^2$ has a jump of size $1/q^2$ (since (qt) jumps by 1 when t crosses p/q). You can show this jump survives in the sum f by checking that the other terms are continuous there or that their jumps don't cancel.

The discontinuities form a countable dense set

Countable: The rationals $\mathbb{Q}$ are countable.

Dense: $\mathbb{Q}$ is dense in $\mathbb{R}$.

Riemann integrability despite the discontinuities

This is the beautiful part: f is discontinuous on a countable dense set, yet still Riemann-integrable!

Key tool: The series converges uniformly (by the M -test with $M_n = 1/n^2$). Each partial sum $S_N(x) = \sum_{n=1}^N (nx)/n^2$ is a finite sum of bounded functions with finitely many discontinuities in any interval, hence each S_N is Riemann-integrable. Theorem 7.16 says: the uniform limit of Riemann-integrable functions is Riemann-integrable.

Proof outline

1. **Uniform convergence:** $|(nx)/n^2| \leq 1/n^2$, Weierstrass M -test.
2. **Discontinuities at rationals:** Show the jump from (nx) at $x = p/q$ doesn't cancel.
3. **Continuity at irrationals:** All terms continuous + uniform convergence $\Rightarrow$ sum continuous (Thm 7.12).
4. **Riemann-integrable:** Each $S_N \in R$, uniform convergence, Thm 7.16.